

Distilled Human Judgment for Public Goods Funding (DHJ)

Aligning LLM Voters to Human Juror Values



Warsaw, 2026

IT - Intelligent Systems and Data Science

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Problem of democratic structures



Avoid concentration of power

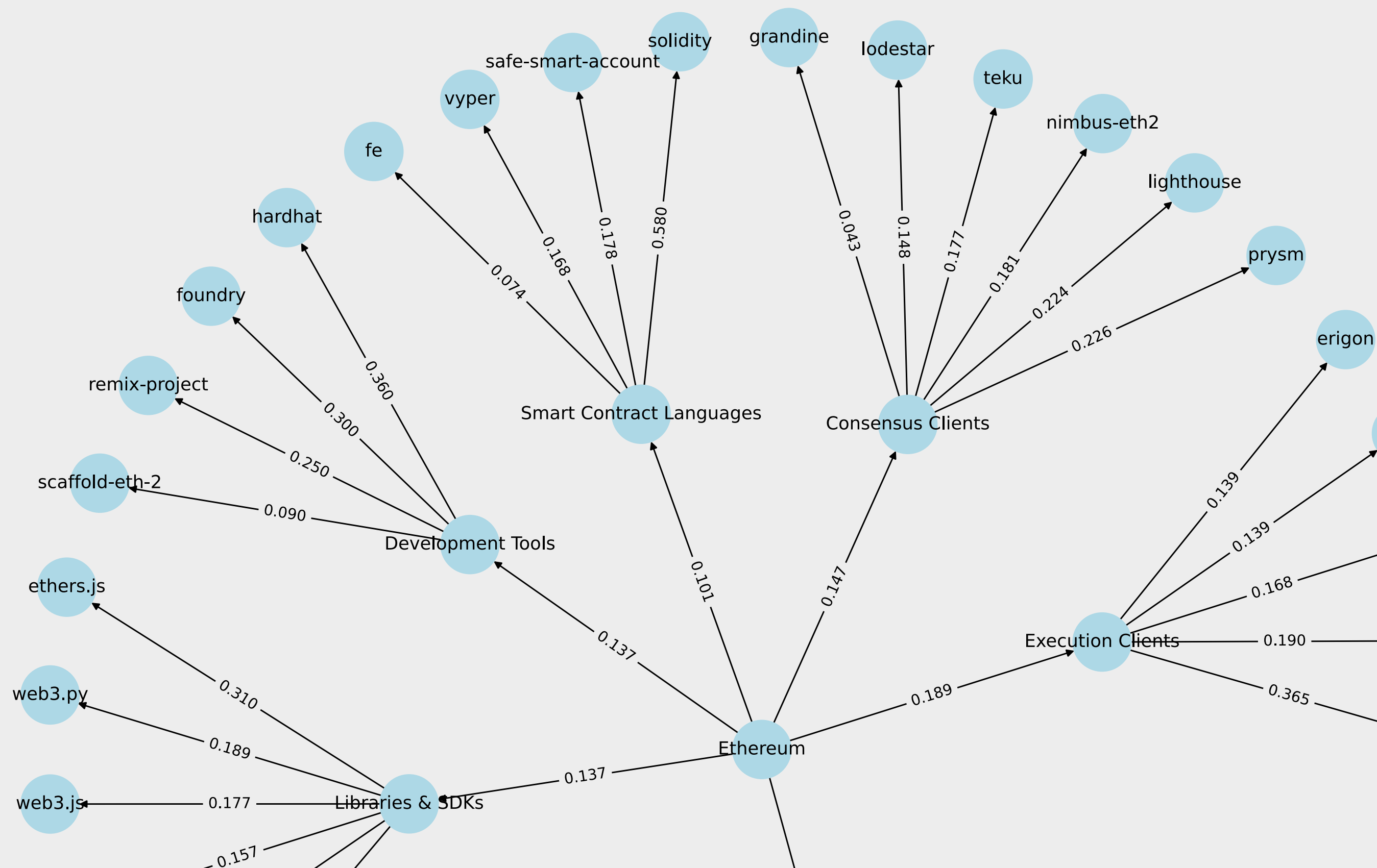


Low participation and low-effort voting in practice



Power recentralizes in a few trusted people and popular voices

Example of such problem



Solution overview:

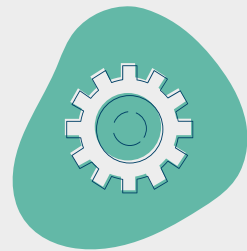
Distilled Human Judgment



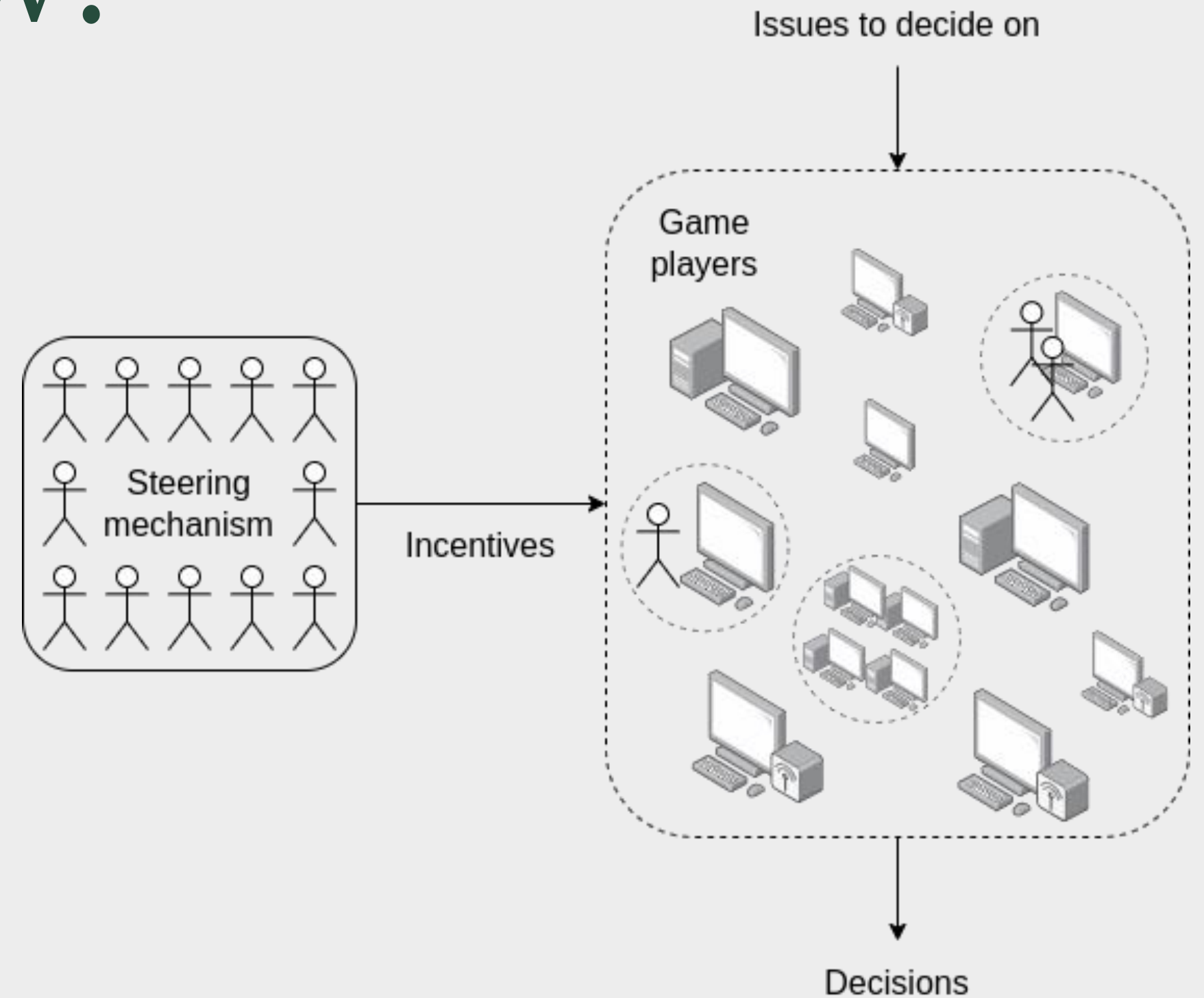
Small jury labels a subset of decisions
(high-quality human signal)



Many AI models score everything
(scale)



Learn an ensemble that best matches
human judgments
(alignment step)



How judgments are represented

Plain text

Humans provide pairwise, multiplicative judgments (e.g., "B is 10× A")

Encode judgments in log-space so ratios become additive

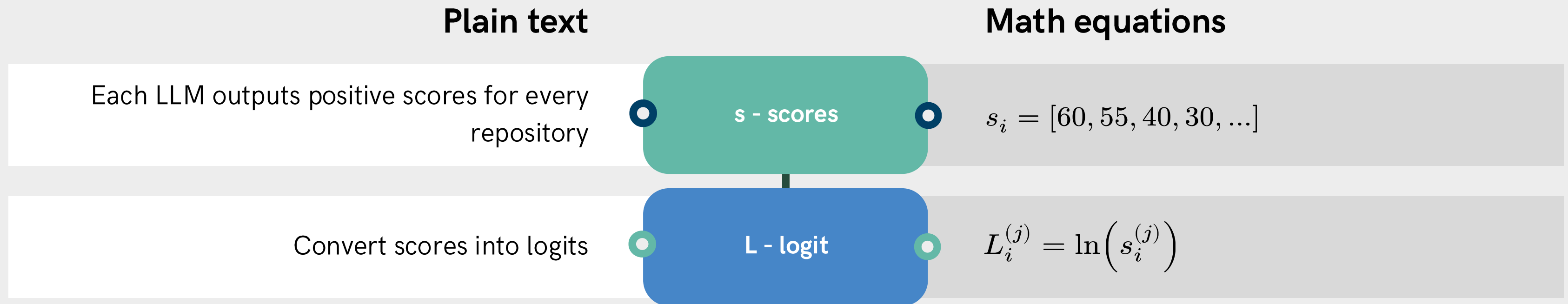
M - multiplier

Math equations

$$c_k = \begin{cases} \ln(M_k) & \text{if } b_k \text{ is preferred} \\ -\ln(M_k) & \text{if } a_k \text{ is preferred} \end{cases}$$

indexes: i models, j repositories, k human-labeled comparisons (a, b)

How LLM scores are represented



indexes: i models, j repositories, k human-labeled comparisons (a, b)

How final scores are produced

Plain text

Math equations

Build a combined logit vector over repositories

$\mathbf{l}(w)$ - logit vector

$$\tilde{L}^{(j)}(w) = \sum_{i=1}^n w_i L_i^{(j)}$$

For each human-labeled comparison between repositories (A, B), the ensemble predicts

$p(w)$ - prediction

$$p_k(w) = \tilde{L}^{(b_k)}(w) - \tilde{L}^{(a_k)}(w)$$

Human labels are signed log-multipliers c. Fit weights by minimizing squared disagreement

w^* - final weights

$$w^* = \arg \min_{w \in \Delta^n} \sum_k (p_k(w) - c_k)^2$$

Combine logit per repository, then exponentiate to get the final score



$$\text{score}^{(j)} = \exp \left(\sum_{i=1}^n w_i^* L_i^{(j)} \right)$$

indexes: i models, j repositories, k human-labeled comparisons (a, b)

Results

Primary metric: Agreement rate

Percent of pairwise comparisons
where LLMs and the jury choose the
same repo

Test set: 96 comparisons

Unique repositories: 47

Jurors: 37

72.92%

vs.

60.42%

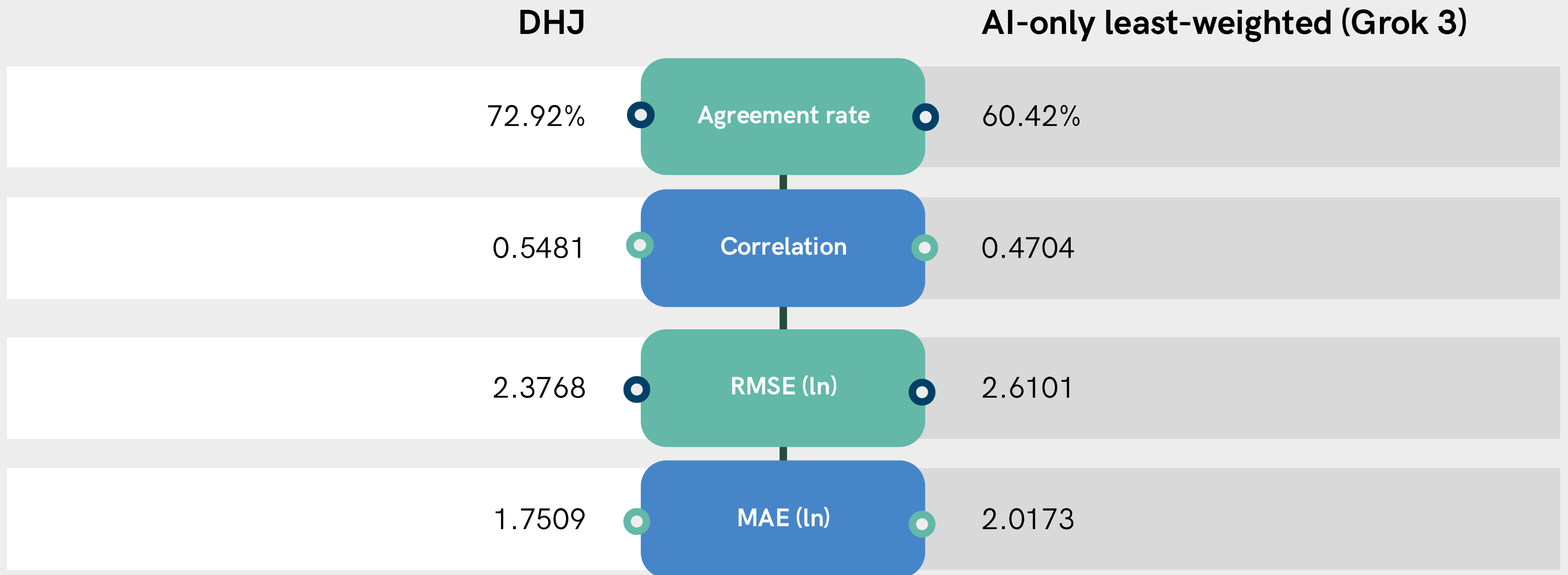
DHJ
(Ensemble of 14 LLMs)

AI-only least-weighted
(Grok 3)



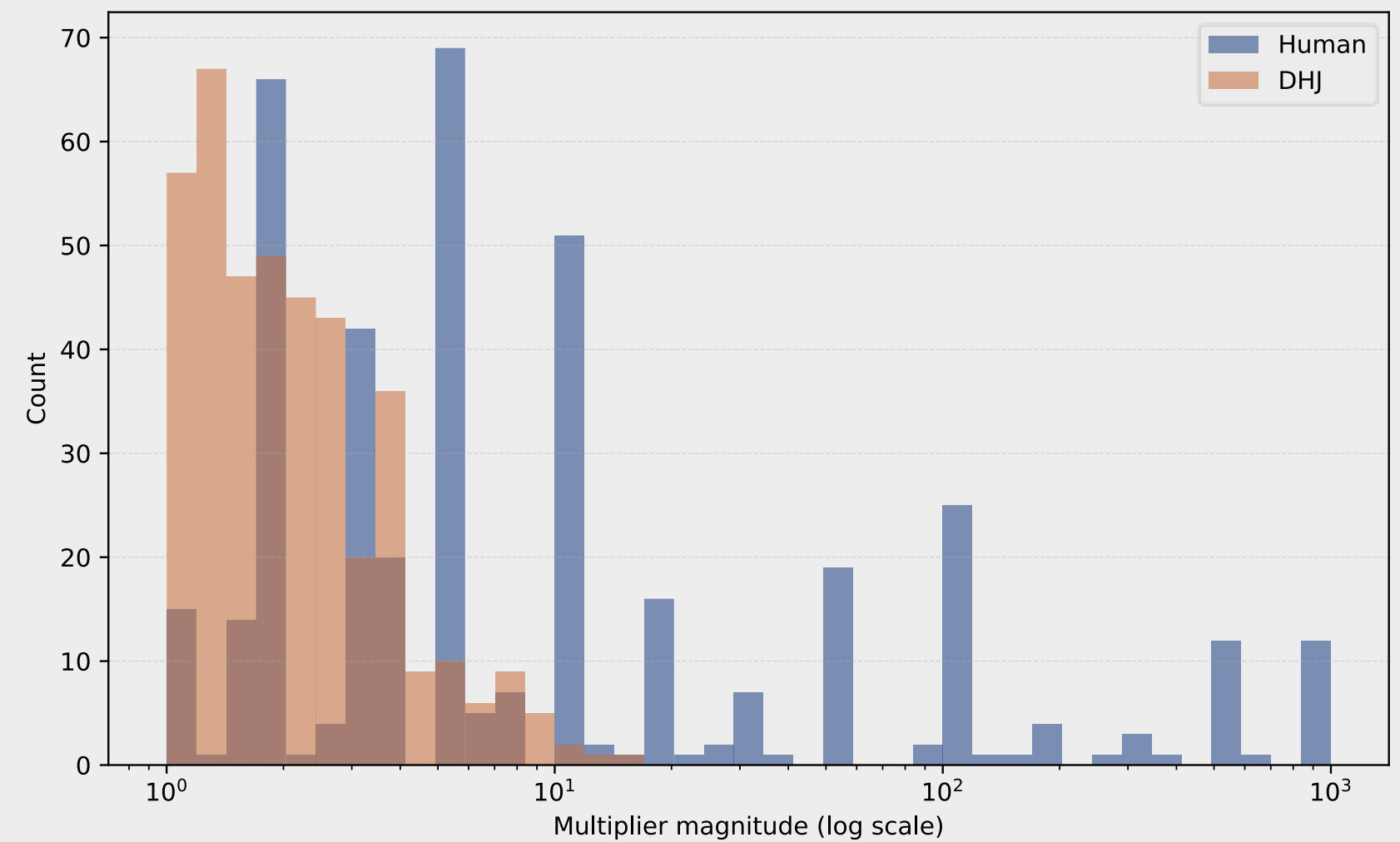
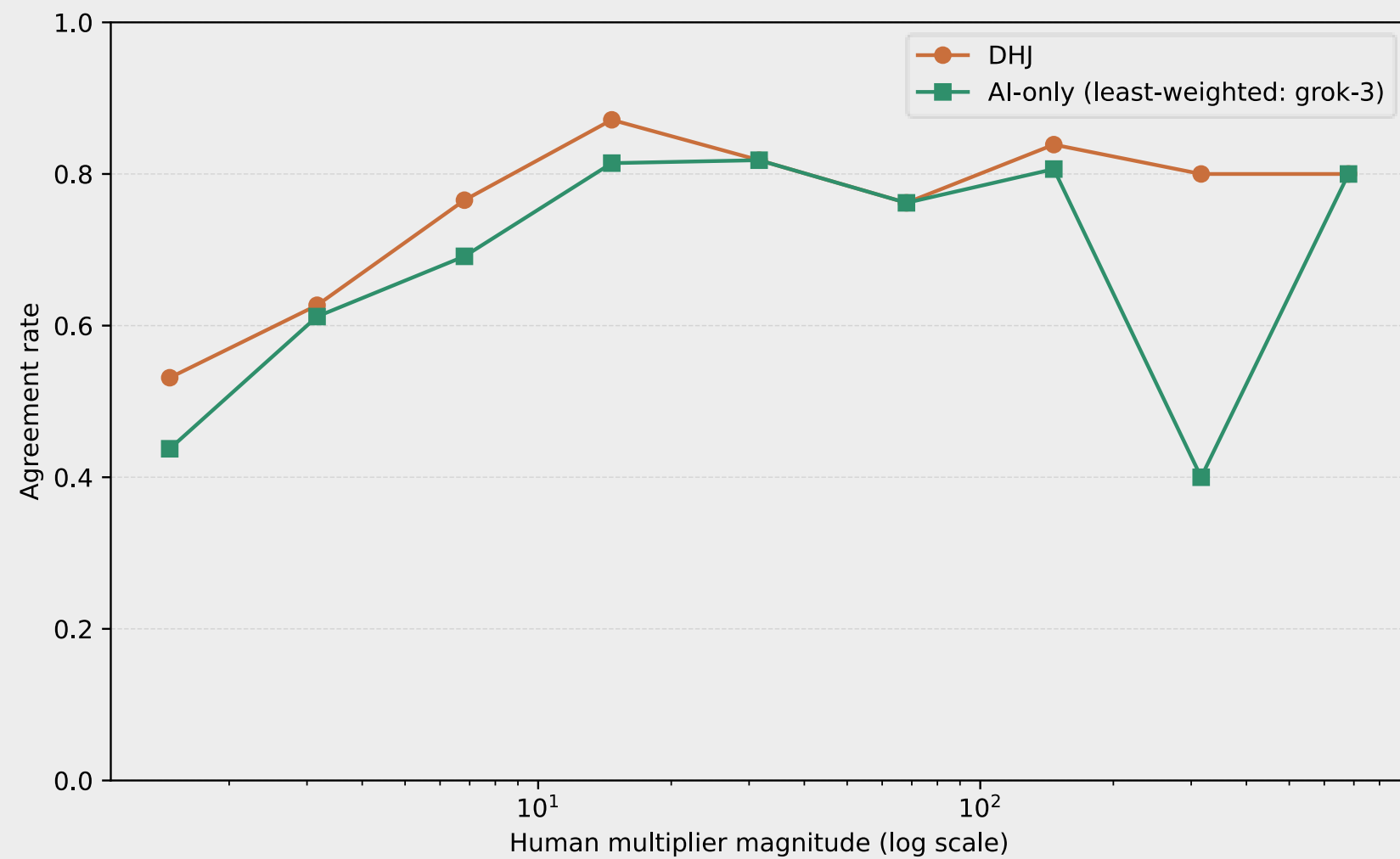
Performance Comparison

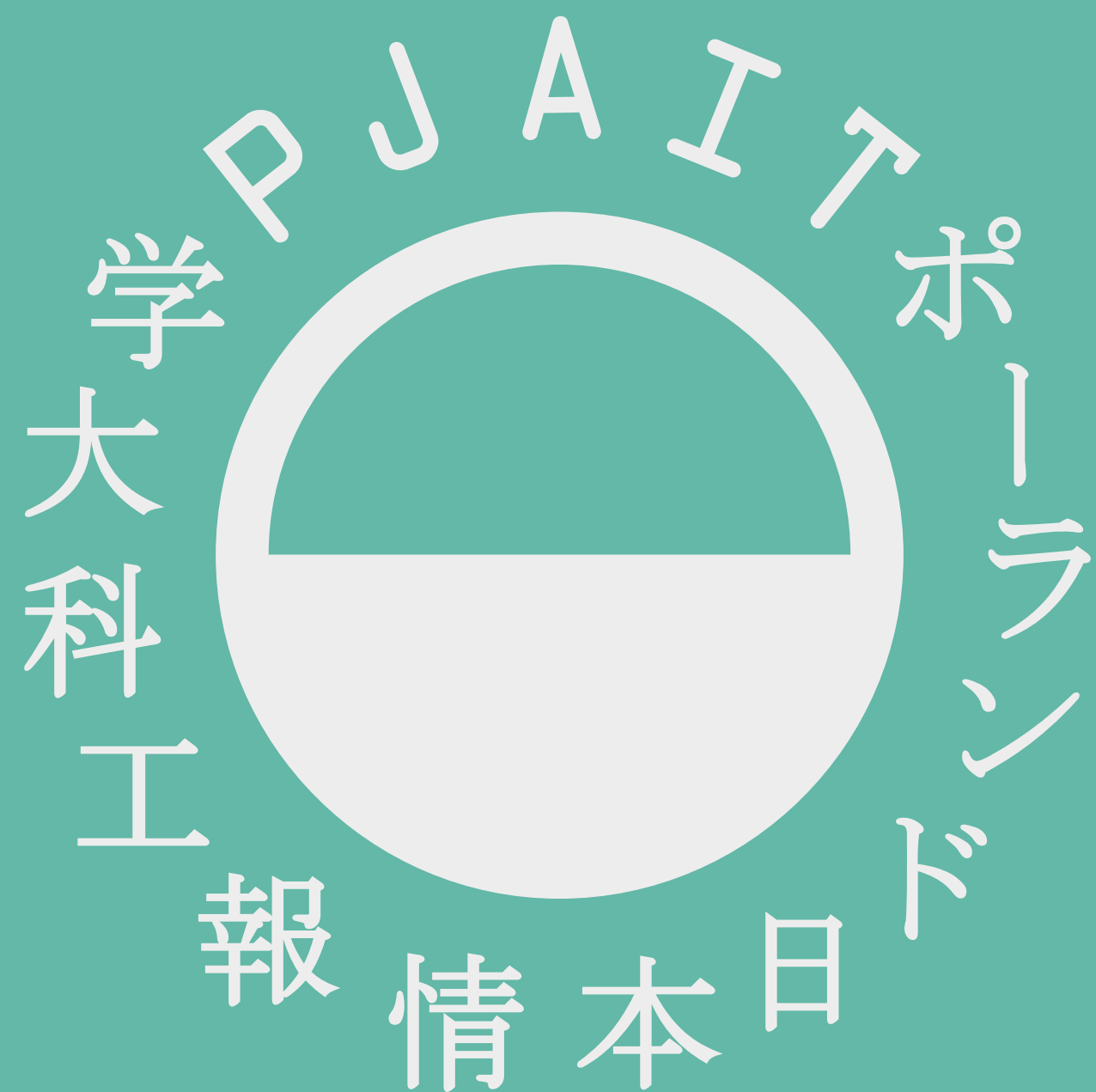
Benchmarking DHJ versus AI-only least-weighted (Grok3) on test-set



Conclusion

Where DHJ succeeds and fails





Thank you!

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